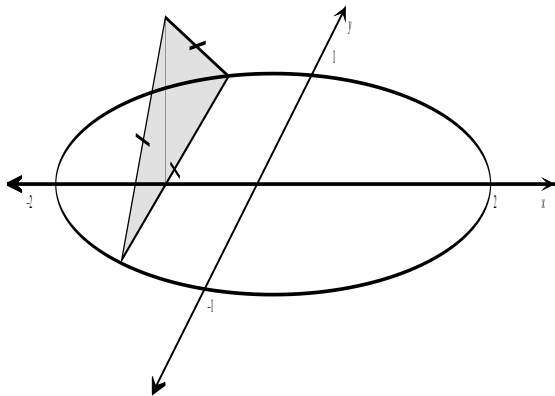


QUESTION 1 (15 Marks)

Marks

(a) A solid shape has as its base an ellipse in the XY plane as shown below. Sections taken perpendicular to the X -axis are equilateral triangles. The major and minor axes of the ellipse are 4 metres and 2 metres respectively.



(i) Write down the equation of the ellipse.

1

(ii) Show that the area of the cross-section at $x = k$ is given by

2

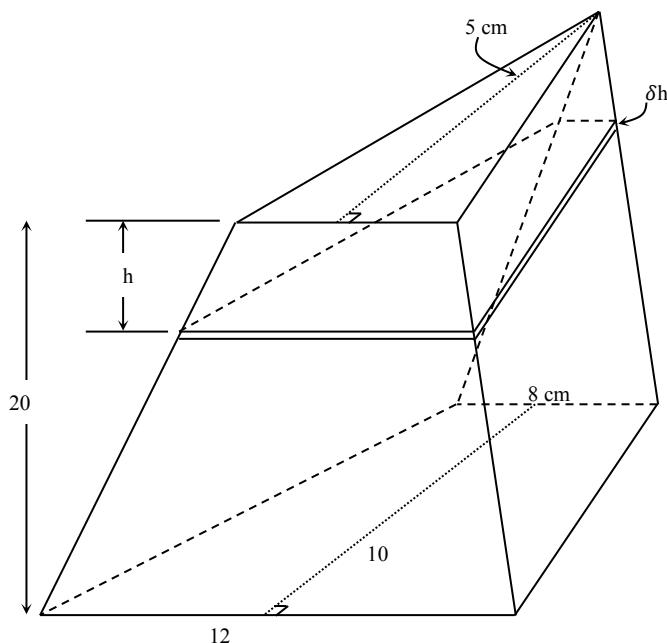
$$A = \frac{\sqrt{3}}{4} (4 - k^2).$$

(iii) By using the technique of slicing, find the volume of the solid.

2

(b)

The solid shown has a base which is a trapezium. The parallel sides are 12cm and 8cm. The perpendicular height is 10cm. Each slice taken parallel to the base is a trapezium with one of the parallel sides 4cm shorter than the other side. The top of the figure (which is parallel to the base) is a triangle with a height of 5cm. The height of the solid is 20cm.



(i) Show that the perpendicular height (w) of the slice shown is given by $w = \frac{h}{4} + 5$.

2

(ii) Find an expression for the volume of the slice shown in the diagram in terms of h .

2

(iii) Find the volume of the solid to 1 decimal place.

2

QUESTION 1 (cont)

- c) (i) Prove that the hyperbola with equation $x^2 - y^2 = a^2$ is the hyperbola $XY = \frac{1}{2} a^2$ referred to different axes. 2
- (ii) Find the coordinates of the vertices, foci and the equations of the directrices of $XY = 4$ 2

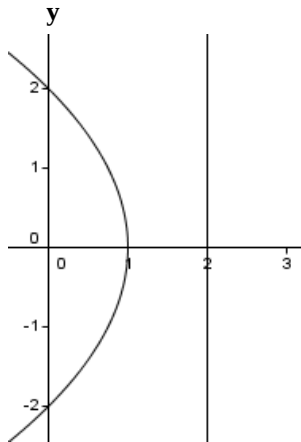
QUESTION 2 Start a NEW PAGE (15 Marks)

- (a) The normal at the point $P\left(cp, \frac{c}{p}\right)$ on the hyperbola $xy=c^2$, meets the x-axis at Q. M is the midpoint of PQ
- (i) Show that the normal at P has the equation $p^3x - py = c(p^4 - 1)$. 2
- (ii) Show that M has the coordinates $\left(\frac{c(2p^4 - 1)}{2p^3}, \frac{c}{2p}\right)$ 2
- (iii) Hence or otherwise, find the equation of the locus of M. 2
- (b) Using the hyperbola from part a) but where $p \neq \pm 1$.
- (i) Write down the equation for the tangent at P. 1
- (ii) If the tangent to the hyperbola at P meets the coordinate axes at A and B. Show that $PA=PB$. 2
- (iii) Let the normal to the hyperbola at P meet the axes of symmetry of the hyperbola at C and D. Show that $PC=PD=PA$. 4
- (iv) Sketch a graph of the hyperbola showing the results for parts so far. 1
- (v) Explain why ACBD is a cyclic quadrilateral and deduce that $BD \perp BC$ 1
- (vi) Describe the geometry if $p=1$ 1

QUESTION 3 Start a NEW PAGE (15 Marks)

Marks

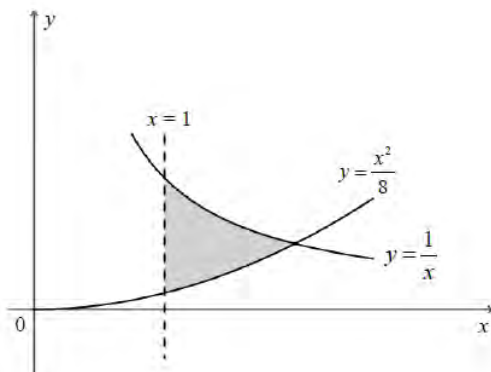
- a) A solid S is formed by rotating the region bounded by the parabola $y^2 = 4(1 - x)$ and the y axis 360° about the line $x = 2$.



Find the volume of the solid S.

4

- (b) The region bounded by $y = \frac{1}{x}$, $y = \frac{x^2}{8}$ and $x = 1$ is rotated about the line $x = 1$.



- (i) Use the method of cylindrical shells to find an integral which gives the volume of the resulting solid of revolution
- (ii) Find the volume of this solid of revolution

3

2

- (c) A solid of mass 2kg is attached to an inextensible string of length 1.5 metres, the other end of the string being fixed. The mass rotates in a horizontal circle with an angular velocity of $\pi \text{ rad s}^{-1}$, forming a conical pendulum. (Take $g = 10 \text{ ms}^{-2}$)

- (i) Calculate the tension in the string.
- (ii) Determine the angle between the string and the vertical axis.
- (iii) Find the radius of the rotation.
- (iv) What is the effect on the motion of the particle if the mass is doubled?

3

1

1

1

QUESTION 4 Start a NEW PAGE (15 Marks)

(a) A particle of mass 2kg is projected vertically upwards from a point A with velocity u m/s. It experiences a resistive force, in Newtons, of 10% of the square of its velocity v metres per second. The highest point reached is B directly above A. Assume $g = 10 \text{ ms}^{-2}$, and take upwards as the positive direction

(i) Show that the acceleration of the particle as it rises is given by $\ddot{x} = -\left(\frac{v^2 + 200}{20}\right)$ 1

(ii) Show that the distance x metres of the particle from A as it rises is given by 3

$$x = 10 \ln \left(\frac{200 + u^2}{200 + v^2} \right)$$

(iii) Show that the time t seconds that the particle takes to reach a velocity of v metres per second is given by $t = \sqrt{2} \left(\tan^{-1} \frac{u}{10\sqrt{2}} - \tan^{-1} \frac{v}{10\sqrt{2}} \right)$ 2

(iv) Now suppose we take two of the 2 kg particles described above. One of the particles is projected upwards from A with an initial velocity $10\sqrt{2} \text{ ms}^{-1}$ then, $\frac{3\sqrt{2}}{5}$ seconds later the other particle is projected upwards from A with initial velocity $20\sqrt{2} \text{ ms}^{-1}$. Will the second particle catch up to the first particle before the first particle reaches its maximum height? You must explain your reasoning and show working. 3

(b) A particle is allowed to fall under gravity from rest in a medium which exerts a resistance proportional to the speed (v) of the particle.

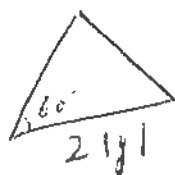
(i) Show that the particle reaches a terminal velocity T given by $T = \frac{g}{k}$ (where k is a positive constant). 2

(ii) Show that the distance fallen to reach half its terminal velocity $\frac{T}{2}$ is given by $x = \frac{T^2}{g} \ln 2 - \frac{T^2}{2g}$. 4

END OF EXAMINATION

i) $\frac{x^2}{4} + y^2 = 1$

ii) $\text{Area} = \frac{1}{2} \cdot 2|y| \cdot 2|y| \sin 60^\circ$
 $= 2y^2 \cdot \frac{\sqrt{3}}{2}$
 $= \sqrt{3}y^2$



When $x=k$ $y^2 = 1 - \frac{k^2}{4}$

$\text{Area} = \sqrt{3} \left(1 - \frac{k^2}{4}\right)$
 $= \frac{\sqrt{3}}{4} (4 - k^2)$

iii) $\Delta V = \frac{\sqrt{3}}{4} (4 - k^2) \Delta k$

$V = \lim_{\Delta k \rightarrow 0} \sum_{k=-2}^2 \frac{\sqrt{3}}{4} (4 - k^2) \Delta k$

$= \int_{-2}^2 \frac{\sqrt{3}}{4} (4 - k^2) dk$

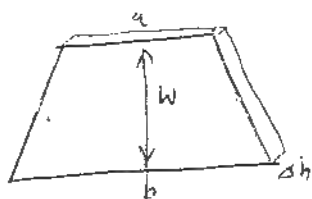
$= 2 \int_0^2 \frac{\sqrt{3}}{4} (4 - k^2) dk$

$= \frac{\sqrt{3}}{2} \left(4k - \frac{k^3}{3}\right) \Big|_0^2$

$= \frac{\sqrt{3}}{2} \left(8 - \frac{8}{3}\right)$

$= \frac{8\sqrt{3}}{3} \text{ or } \frac{8}{\sqrt{3}} \text{ unit}^3$

b i)



$w(h)$ is a linear function

$w(h) = mh + c$

when $h=0$ $w=5$ $\therefore c=5$
 $h=20$ $w=10$

$10 = 20m + 5$ $m = \frac{1}{4}$

$\therefore w = \frac{h}{4} + 5$

1 m well done

1 m

Some forgot $x=k$
 $-\frac{1}{2}m$

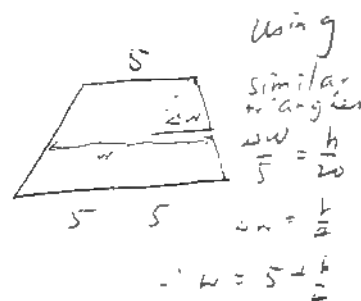
1 m

If use

$V = 2 \int_0^2 \frac{\sqrt{3}}{4} (4 - k^2) dk$

must mention even function
 or symmetrical

1 m



1 m

Some students

assume symmetrical
 or right-angled
 trapezium -1 m

must mention

similar triangles

1 m

Similarly

$$a(h) = mh + b$$

When $h=0$ $a=0 \rightarrow b=0$
 $h=20$ $a=8$

$$8 = 20m \quad \therefore m = \frac{2}{5}$$

$$\therefore a = \frac{2h}{5}$$

$$b = \frac{2h}{5} + 4$$

$$\Delta V = \frac{1}{2} \left(\frac{h}{4} + 5 \right) \left[\frac{2h}{5} + \frac{2h}{5} + 4 \right] \Delta h$$

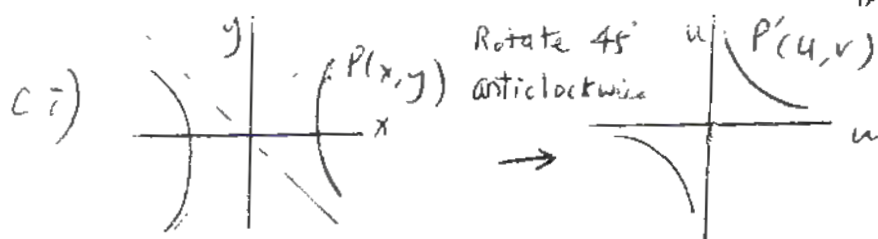
$$V = \lim_{\Delta h \rightarrow 0} \sum_{h=0}^{\infty} \frac{1}{2} \left(\frac{h}{4} + 5 \right) \left(4 + \frac{4h}{5} \right) \Delta h$$

$$= \frac{1}{2} \int_0^{20} \left(\frac{h}{4} + 5 \right) \left(4 + \frac{4h}{5} \right) dh$$

$$= \int_0^{20} \left(\frac{h^2}{10} + \frac{h}{2} + 2h + 10 \right) dh$$

$$= \left(\frac{h^3}{30} + \frac{5h^2}{4} + 10h \right) \Big|_0^{20}$$

$$= \frac{2900}{3} \text{ cm}^3 \approx 966.7 \text{ cm}^3 \quad \#$$



$P(x, y)$ represents $z = x\hat{i} + y\hat{j}$, $P'(u, v)$ represents $(\text{cis } 45^\circ) \cdot z$

$$u + iv = \frac{1}{\sqrt{2}} (1 + i)(x + iy) = \left(\frac{x-y}{\sqrt{2}} \right) + i \left(\frac{x+y}{\sqrt{2}} \right)$$

$$u = \frac{x-y}{\sqrt{2}}, \quad v = \frac{x+y}{\sqrt{2}}$$

$$uv = \frac{x^2 - y^2}{2} = \frac{a^2}{2} \quad \left(\text{since } P \text{ lies on } x^2 - y^2 = a^2 \right)$$

Similarly

The front & rear sides may not be symmetrical

/m

/m

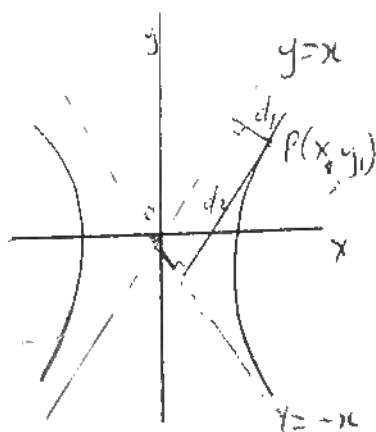
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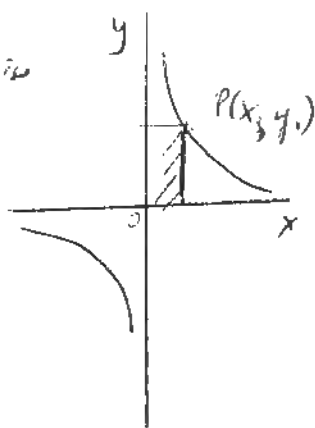
/m

(i) $x^2 - y^2 = a^2$



Rotate
45°
anticlockwise

$xy = \frac{a^2}{2} (=c^2)$



$$d_1 = \frac{|x_1 - y_1|}{\sqrt{2}}$$

$$d_2 = \frac{|x_1 + y_1|}{\sqrt{2}}$$

$$\text{Area of rectangle} = \frac{|x_1^2 - y_1^2|}{2} = \frac{a^2}{2}$$

(Since $P(x_1, y_1)$ lies
on $x^2 - y^2 = a^2$)

After an anticlockwise rotation of 45°, area of rectangle is still $\frac{a^2}{2}$ (Now the x, y axes become the new asymptote)

$$\therefore x_1 y_1 = \frac{a^2}{2}$$

$\therefore$ the hyperbola with equation $x^2 - y^2 = a^2$ is the hyperbola $xy = \frac{a^2}{2}$ referred to different axes

(ii) Vertices $(2, 2), (-2, -2)$

Foci $(2\sqrt{2}, 2\sqrt{2}), (-2\sqrt{2}, -2\sqrt{2})$

Directrices $x + y = \pm 2\sqrt{2}$

Using this method
must mention
equation of
asymptotes
otherwise - 1 m.

1 m

1 m

1 m for 1 correct
2 m for all 3 correct
no half mark.

1/4

MATHEMATICS Extension 2: Question 2

Suggested Solutions	Marks	Marker's Comments
(i) $xy = c^2 \therefore y = \frac{c^2}{x}$ $\frac{dy}{dx} = -\frac{c^2}{x^2}$ when $x = cp$, $m = \frac{-c^2}{c^2 p^2} = -\frac{1}{p}$ gradient of normal is p^2 eqn of normal is $y - \frac{c}{p} = p^2(x - cp)$ $py - c = p^3 x - cp^4$ $cp^4 - c = p^3 x - py$ $c(p^4 - 1) = p^3 x - py$	① ①	generally very well done.
(ii) when $y = 0$, $c(p^4 - 1) = p^3 x$ $x = \frac{c(p^4 - 1)}{p^3}$ $\left(\frac{c(p^4 - 1)}{p^3}, 0 \right)$ Midpoint is $\left(\frac{\frac{c(p^4 - 1)}{p^3} + cp}{2}, \frac{0 + \frac{c}{p}}{2} \right)$ $= \left(\frac{c(p^4 - 1) + cp^4}{2p^3}, \frac{c}{2p} \right)$ $= \left(\frac{2cp^4 - c}{2p^3}, \frac{c}{2p} \right)$	① ①	
(iii) $x = \frac{2cp^4 - c}{2p^3}$ $y = \frac{c}{2p}$ $\therefore p = \frac{c}{2y}$ sub into x coordinate $\therefore x = \frac{2c\left(\frac{c}{2y}\right)^4 - c}{2\left(\frac{c}{2y}\right)^3} = \frac{c}{2} - \frac{c}{2p^3}$ $\therefore x = \frac{c}{2} - \frac{c}{2\left(\frac{c}{2y}\right)^3} = \frac{c}{2} - \frac{c}{2\left(\frac{c^3}{8y^3}\right)}$	①	

2/4

MATHEMATICS Extension 2: Question... 2.

Suggested Solutions	Marks	Marker's Comments
$\therefore x = \frac{c^2}{2y} - \frac{8y^3}{2c}$ $\therefore x = \frac{c^2}{2y} - \frac{4y^3}{c^2}$ $\text{or } xc^2 = \frac{c^4}{2y} - 4y^3 \quad \text{or } 2yx^2 = c^4 - 8y^4$	①	you had to substitute into x, and do at least 2 further lines of working to get ②
(b)(i) $x + p^2y = 2cp$ (ii) when $x=0$, $p^2y = 2cp$ so $A(0, \frac{2c}{p})$ when $y=0$, $x = 2cp$ so $B(2cp, 0)$ midpoint of AB is $(\frac{0+2cp}{2}, \frac{2cp+0}{2})$ $= (cp, \frac{2cp}{2})$ $= (cp, \frac{c}{p})$ $\therefore P$ is the midpoint of AB.	①	a lot of students wasted time by proving this formula needed to get both correct to get the 1 mark.
(iii) $y=x$ and $p^3x - py = c(p^4 - 1)$ $p^3x - px = c(p^4 - 1)$ $x(p^3 - p) = c(p^4 - 1)$ $x = \frac{c(p^4 - 1)}{p^3 - p}$ $= \frac{c(p^2 - 1)(p^2 + 1)}{p(p^2 - 1)}$ $= \frac{c(p^2 + 1)}{p}$ $\therefore (\frac{c(p^2 + 1)}{p}, \frac{c(p^2 + 1)}{p})$	①	some students did it by using the distance formula.
$y=-x$ and $p^3x - py = c(p^4 - 1)$ $p^3x + px = c(p^4 - 1)$ $x(p^3 + p) = c(p^4 - 1)$ $x = \frac{c(p^4 - 1)}{p(p^2 + 1)} = \frac{c(p^2 - 1)}{p}$	①	Some students didn't simplify the algebra, and that made it harder for them in applying it.

$$\therefore D \left(\frac{c(p^2 - 1)}{p}, -\frac{c(p^2 - 1)}{p} \right)$$

①

3/4

MATHEMATICS Extension 2: Question 2

Suggested Solutions

Marks

Marker's Comments

midpoint of CD is $\left(\frac{c(p^2+1)}{p}, \frac{c(p^2-1)}{p} \right)$

$$= \left(\frac{2cp^2}{2p}, \frac{2c}{2p} \right)$$

$$= (cp, c/p)$$

$\therefore P$ is the midpoint of CD

now $PC = \sqrt{\left(cp - \frac{c(p^2+1)}{p} \right)^2 + \left(\frac{c}{p} - \frac{c(p^2-1)}{p} \right)^2}$

$$= \sqrt{\left(\frac{cp^2 - cp^2 - c}{p} \right)^2 + \left(\frac{c - cp^2 + c}{p} \right)^2}$$

$$= \sqrt{\frac{c^2}{p^2} + \frac{c^2 p^4}{p^2}}$$

$$= \frac{c}{p} \sqrt{1 + p^2}$$

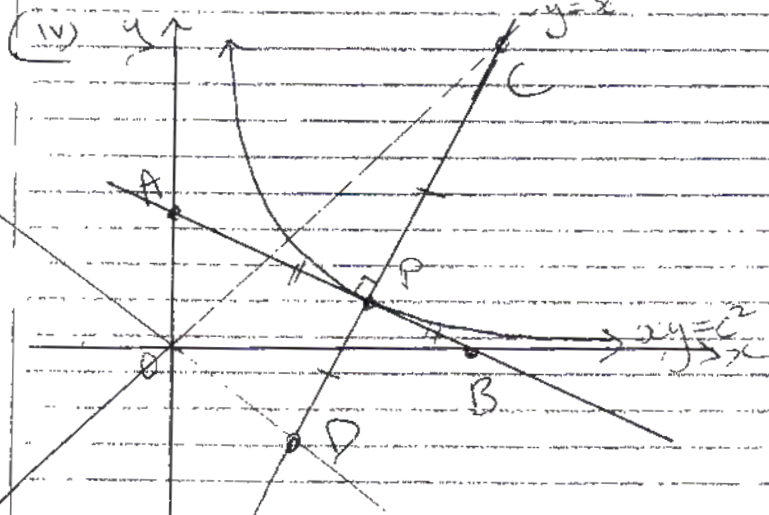
$$PA = \sqrt{(cp - c)^2 + \left(\frac{c}{p} - \frac{2c}{p} \right)^2}$$

$$= \sqrt{c^2 p^2 + \frac{c^2}{p^2}}$$

$$= \frac{c}{p} \sqrt{p^4 + 1}$$

$$= PC$$

$\therefore PA = PC = PD$ (as P midpoint of CD)

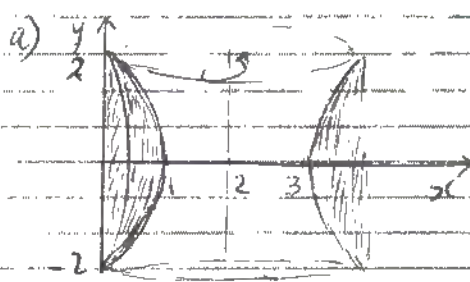
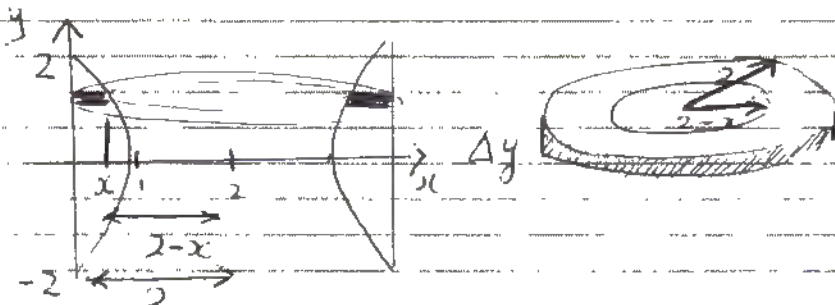


①
needed to
see A, B, C, D, P
and $y = \pm x$
to get the 1

MATHEMATICS Extension 2: Question 2.....

Suggested Solutions	Marks	Marker's Comments
(v) The diagonals AB and CD bisect each other at right angles, $AB = CD$ so $ACBD$ is a square. It is a cyclic quadrilateral as the opposite angles are supplementary. $BD \perp BC$ (angle B in a square is 90°)		* some students forgot to prove why $B = 90^\circ$
(vi) If $p=1$, the normal becomes the axis of symmetry, i.e. $y=x$. So C has infinite values and D goes to the origin.	①	you needed all 3 pieces of information to get the mark. 2 out of 3 = 0ml

EXTENSION 2.
MATHEMATICS: Question... 3

Suggested Solutions	Marks	Marker's Comments
a)  $y^2 = 4(1-x)$ 2 methods <u>Method 1 SLICES.</u>  $y^2 = 4(1-x)$ $x = \frac{4-y^2}{4} = 1 - \frac{y^2}{4}$ $\Delta V \equiv \pi [2^2 - (2-x)^2] \Delta y$ $\equiv \pi [4 - 4 + 4x - x^2] \Delta y$ $\equiv \pi x [4-x] \Delta y$ $\equiv \pi \left[\frac{4-y^2}{4} \right] \left[4 - \frac{4-y^2}{4} \right] \Delta y$ $\equiv \pi \left[\frac{4-y^2}{4} \right] \left[\frac{12+y^2}{4} \right] \Delta y$ $\equiv \frac{\pi}{16} [48 - 8y^2 - y^4] \Delta y$ $V = \lim_{\Delta y \rightarrow 0} \left\{ \frac{\pi}{16} [48 - 8y^2 - y^4] \right\}$ $= \frac{\pi}{16} \int_{-2}^2 (48 - 8y^2 - y^4) dy$ $= \frac{2\pi}{16} \int_0^2 (48 - 8y^2 - y^4) dy \quad (\text{even function})$ $= \frac{\pi}{8} \left[48x - \frac{8y^3}{3} - \frac{y^5}{5} \right]_0^2$ $= \frac{128\pi}{15}$ Volume is $\frac{128\pi}{15}$ units³.	④	4 For complete answer. No marks deducted for missing explanation or working ① ΔV statements ① sub y into ΔV ① simplify algebra in integral ① integrate and answer.

EXTENSION
MATHEMATICS: Question...

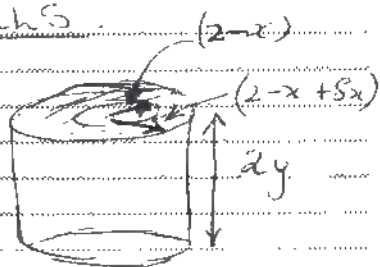
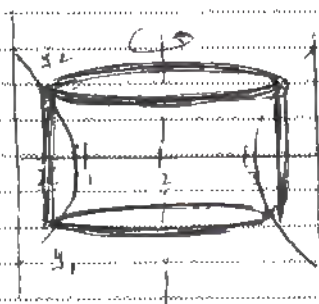
Suggested Solutions

Marks

Marker's Comments

Method 2

CHEHNS

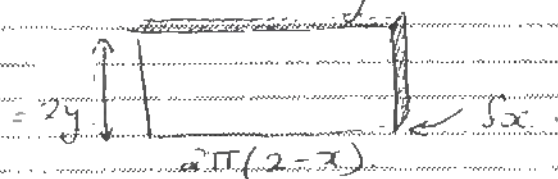


$$\Delta V = \pi [(2-x+\delta x)^2 - (2-x)^2] \times 2y$$

or small cylinders

$$y = 4\sqrt{1-x}$$

$$y = 2\sqrt{1-x} \quad y > 0$$



$$\Delta V = 2\pi(2-x) \times 2y \delta x$$

$$= 4\pi(2-x) \times 2\sqrt{1-x} \delta x$$

$$V = \lim_{\delta x \rightarrow 0} 8\pi \sum_{x=0}^1 (2-x) \sqrt{1-x} \delta x$$

$$= 8\pi \int_0^1 (2-x) \sqrt{1-x} dx$$

Use substitution $u = 1-x$ $du = -dx$

$x=1$ $u=0$

$x=0$ $u=1$

$$V = -8\pi \int_1^0 (1+u) \sqrt{u} du$$

$$= 8\pi \int_0^1 u^{\frac{1}{2}} + u^{\frac{3}{2}} du$$

$$= 8\pi \left[\frac{2}{3} u^{\frac{3}{2}} + \frac{2}{5} u^{\frac{5}{2}} \right]_0^1$$

$$= \frac{128\pi}{15}$$

Volume is $\frac{128\pi}{15} \text{ units}^3$

① ΔV statement

① Sub x into ΔV to form integral

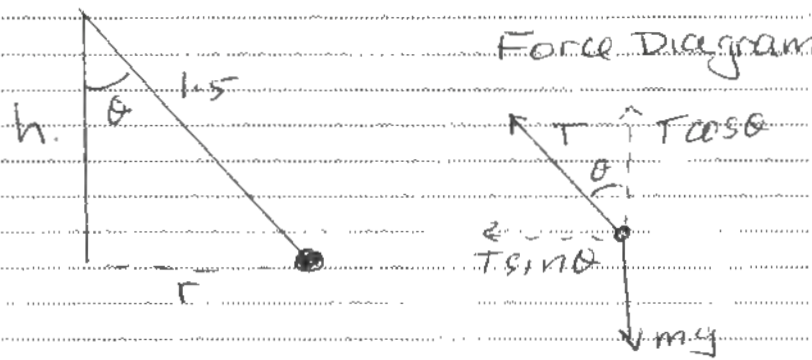
① complete integration

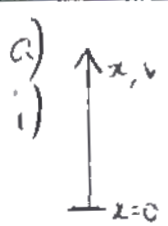
① complete answer

EXTENSION
MATHEMATICS: Question 3...

Suggested Solutions	Marks	Marker's Comments
(b) $y = \frac{x^2}{8}$ $y = \frac{1}{x}$ Point of intersection $\frac{x^2}{8} = \frac{1}{x} \Rightarrow x^3 = 8 \Rightarrow x = 2$ $\Delta V = \pi [(x-1+\delta x)^2 - (x-1)^2] \delta x$ or $\Delta V = 2\pi (x-1) (y_2 - y_1) \delta x$ $y_2 = \frac{1}{x} \quad y_1 = \frac{x^2}{8}$ $\Delta V = 2\pi (x-1) \left[\frac{1}{x} - \frac{x^2}{8} \right] \delta x$ $V = \lim_{\delta x \rightarrow 0} 2\pi \sum_{x=1}^2 (x-1) \left(\frac{1}{x} - \frac{x^2}{8} \right) \delta x$ $V = 2\pi \int_1^2 \left(1 - \frac{1}{x} - \frac{x^3}{8} + \frac{x^2}{8} \right) dx$ (ii) $V = 2\pi \left[x - \ln x - \frac{x^4}{32} + \frac{x^3}{24} \right]_1^2$ $= 2\pi \left[2 - \ln 2 - \frac{16}{32} + \frac{8}{24} \right] - \left[1 - \ln 1 - \frac{1}{32} + \frac{1}{24} \right]$ $= 2\pi \left[\frac{79}{96} - \ln 2 \right]$ Volume = $\left[\frac{79}{48} - 2 \ln 2 \right] \pi \text{ units}^3$	(3)	① point of intersection x value and limits on integral ① diagrams showing variables and ΔV statement ① limit statement and correct integral (any form). ① integration ① answer. no cfe if answer does not contain $\ln$ term.

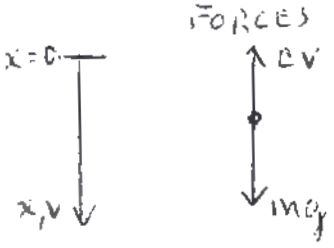
EXTENSION 2
MATHEMATICS: Question 3

Suggested Solutions	Marks	Marker's Comments
c)  Force Diagram (i) Horizontally $T \sin \theta = m \omega^2 r$ $m = 2 \quad \omega = \pi \quad g = 10$ $r = 1.5 \sin \theta$ $T \sin \theta = 2 \times \pi^2 \times 1.5 \sin \theta$ $T = 3\pi^2$ Tension = $3\pi^2$ Newton. (ii) Vertically $T \cos \theta = mg$ $3\pi^2 \cos \theta = 2 \times 10$ $\cos \theta = \frac{20}{3\pi^2}$ $\theta = \cos^{-1}\left(\frac{20}{3\pi^2}\right) \approx 47.31^\circ$ (iii) $r = 1.5 \sin \theta$ $= 1.5 \sin \left[\cos^{-1}\left(\frac{20}{3\pi^2}\right) \right]$ ≈ 1.11 radius = 1.11 m Alternatively $r = \frac{1}{2\pi^2} \sqrt{9\pi^4 - 400}$ (iv) $T \sin \theta = m \omega^2 r = m \omega^2 l \sin \theta$ $T \cos \theta = mg$ $\cos \theta = \frac{mg}{m \omega^2 l} = \frac{g}{\pi^2 l}$ $\therefore \cos \theta$ is independent of m. $\therefore$ no change to motion for $\omega = \pi$	(3) ① Horizontal Equation ① $r = 1.5 \sin \theta$ ① Answer ① ① correct answer. (exact form accepted) ① ① correct answer (Exact form accepted) ① ① no change to motion (no justification needed).	

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a)  i) $mg + 0.1v^2 = 20 + 0.1v^2$ with given values $m\ddot{x} = \sum F_{up}$ (Newton's 2nd Law) $2\ddot{x} = -(20 + 0.1v^2)$ $\ddot{x} = -\frac{(20 + 0.1v^2)}{2}$ $\ddot{x} = -\left(\frac{200 + v^2}{20}\right)$ ii) $\ddot{x} = v \frac{dv}{dx} = -\left(\frac{v^2 + 200}{20}\right)$ $\int_u^v \frac{v dv}{v^2 + 200} = -\int_0^x \frac{dx}{20}$ $\left[\frac{1}{2} \ln(v^2 + 200)\right]_u^v = -\left[\frac{x}{20}\right]_0^x$ $\ln(v^2 + 200) - \ln(u^2 + 200) = -x/10$ $x = 10(\ln(u^2 + 200) - \ln(v^2 + 200))$ $= 10 \ln\left(\frac{200 + u^2}{200 + v^2}\right)$ iii) $\ddot{x} = \frac{dv}{dt} = -\left(\frac{v^2 + 200}{20}\right)$ $\int_u^v \frac{dv}{v^2 + 200} = -\int_0^t \frac{dt}{20}$	1 1 1 1	Given result Show needed either a clear diagram or words "Newton's 2nd Law" to explain the equation of motion. Most people scored full marks here. A few used $\frac{d}{dx}\left(\frac{v^2}{2}\right)$ and got into trouble with limits. There was a temptation to jump to the given result.

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$\left[\frac{1}{\sqrt{200}} \tan^{-1} \frac{v}{\sqrt{200}} \right]_u^v = - \left[\frac{t}{20} \right]_0^t$ $\frac{1}{10\sqrt{2}} \left(\tan^{-1} \frac{v}{10\sqrt{2}} - \tan^{-1} \frac{u}{10\sqrt{2}} \right) = - \frac{t}{20}$ $\therefore t = \frac{20}{10\sqrt{2}} \left(\tan^{-1} \frac{u}{10\sqrt{2}} - \tan^{-1} \frac{v}{10\sqrt{2}} \right)$ $t = \sqrt{2} \left(\tan^{-1} \frac{u}{10\sqrt{2}} - \tan^{-1} \frac{v}{10\sqrt{2}} \right)$	1	Nearly all were able to complete this
iv) The first particle reaches a max ht. of $x_{\max} = 10 \ln \left(\frac{200 + (10\sqrt{2})^2}{-200} \right) = \underline{\underline{10 \ln 2}} \text{ (m)}$ after a time given by $t = \sqrt{2} \left(\tan^{-1} \frac{10\sqrt{2}}{10\sqrt{2}} - \tan^{-1} 0 \right)$ $= \sqrt{2} \tan^{-1}(1) = \underline{\underline{\frac{\sqrt{2}\pi}{4}}} \text{ (secs)}$	1	First mark for getting <u>both</u> of these values
For second particle there are two approaches: Ⓐ Find where 2nd particle is When $t = \frac{\pi\sqrt{2}}{4} - \frac{3\sqrt{2}}{5}$ $t \rightarrow v \rightarrow x$ Compare heights at this time Ⓑ Find time taken to reach $x = 10 \ln 2$ $x \rightarrow v \rightarrow t$ Compare times taken (include offset)		

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(A) Find v when $t = \frac{\pi\sqrt{2}}{4} - \frac{3\sqrt{2}}{5}$ $\therefore \frac{\pi\sqrt{2}}{4} - \frac{3\sqrt{2}}{5} = \sqrt{2} \left(\tan^{-1} \frac{20\sqrt{2}}{10\sqrt{2}} - \tan^{-1} \frac{v}{10\sqrt{2}} \right)$ $\tan^{-1} \left(\frac{v}{10\sqrt{2}} \right) = \tan^{-1}(2) + \frac{3}{5} - \frac{\pi}{4}$ $= 0.92175 \dots$ $\therefore v = 10\sqrt{2} \tan(0.92175 \dots)$ $= 18.63996 \dots$ Thence $x = 10 \ln \left(\frac{200 + (20\sqrt{2})^2}{200 + (18.63996)^2} \right)$ $= 6.025 \quad (3DP)$ $< 10 \ln 2 (= 6.931)$ <u>$\therefore$ Particle does not pass first in this time range.</u>	1	Rather too many people used their calculator in Degrees mode - not a good idea.
(B) Find v when $x = 10 \ln 2$ $10 \ln 2 = 10 \ln \left(\frac{200 + (20\sqrt{2})^2}{200 + v^2} \right)$ $2 = \frac{1000}{200 + v^2}$ $v^2 = 300, \quad v = 10\sqrt{3} \text{ (up)}$ Time since takeoff, T, given by $T = \sqrt{2} \left(\tan^{-1} \left(\frac{20\sqrt{2}}{10\sqrt{2}} \right) - \tan^{-1} \left(\frac{10\sqrt{3}}{10\sqrt{2}} \right) \right)$ $= 0.31264 \dots$ Time since $t=0$ is $T + \frac{3\sqrt{2}}{5}$ $= 1.16117 \dots$ $> \frac{\pi\sqrt{2}}{4} (= 1.1107 \dots)$ <u>$\therefore$ Particle 2 does not catch first before top point</u>	1	

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b) i) $x=0$  $m\ddot{x} = mg - cv$ $\therefore \ddot{x} = g - kv$ where $k = c/m > 0$ As $x \rightarrow 0$, $v \rightarrow T$ $g - kT = 0$ $T = g/k$	1	Should be an easy 2 marks but there was much fudging of m.
ii) $\therefore v \frac{dv}{dx} = g - kv$ $\int_0^{T/2} \frac{v dv}{g - kv} = \int_0^x dx$ $\frac{1}{k} \int_0^{T/2} \frac{kv - g + g}{g - kv} dv = \int_0^x dx$ $\frac{1}{k} \int_0^{T/2} \left(\frac{g}{g - kv} - 1 \right) dv = x$ $\therefore \left[-\frac{1}{k^2} \ln g - kv - \frac{v}{k} \right]_0^{T/2} = x$ $0 < v < g/k$ so abs. values not nec. $-\frac{1}{k^2} \ln\left(\frac{g - kT/2}{g}\right) - \frac{T}{2k} = x$ But $k = g/T$ from (i) $\therefore x = -\frac{T^2}{g^2} \ln\left(\frac{g - g/2}{g}\right) - \frac{T^2}{2g}$ $= \frac{T^2}{g^2} \ln 2 - \frac{T^2}{2g}$	1 1 1 1	One mark for dealing inadequately with absolute value signs. Total omission of them, without explanation, lost one mark. Given result, g need an extra line before end for $\ln 2$